SQL*Plus Setup & Database Connection Guide

Windows • Oracle Autonomous Database • Wallet-based connection

Purpose
Install Oracle SQL*Plus, configure the Autonomous Database wallet, and connect to the training database from a Windows command prompt.

Example service alias
`sqlworkshop_medium`

Participant setup handout

Security reminder: Use the credentials and wallet supplied by your instructor or administrator. Never publish passwords or wallet files in a public repository.

1. What you need before starting

Confirm that you have the following items:

Requirement	What to check
Windows computer	You can open Command Prompt and edit user environment variables.
Oracle SQL*Plus	SQL*Plus executable compatible with your Windows architecture (normally 64-bit).
Instant Client Basic package	Required runtime libraries. Use the Basic and SQL*Plus packages from the same version.
Wallet ZIP	The Autonomous Database wallet ZIP supplied for the training database.
Database credentials	Database username and password, provided separately by the instructor or administrator.

How the pieces fit together: SQL*Plus is the command-line client. The wallet and network configuration allow it to reach Autonomous Database securely. Your database username and password authenticate your account.

2. Download and install SQL*Plus

1. Open Oracle's official Instant Client Downloads page.

2. Select the Windows 64-bit packages unless your environment requires another architecture.

3. Download both **Basic** (or Basic Light, if suitable) and **SQL*Plus** ZIP packages for the **same version**.

4. Create a folder such as:

```
C:\oracle\instantclient_23_x
```

Replace the example version folder with the folder name you actually downloaded.

5. Extract the contents of both ZIP packages into the same Instant Client folder. The folder should contain `sqlplus.exe` and Oracle DLL files.

6. Add the Instant Client folder to Windows **Path**: Start → search for “Edit environment variables for your account” → select **Path** → Edit → New → enter the folder path → OK.

7. Close open Command Prompt windows, open a new one, and verify the installation:

```
sqlplus -v
```

A SQL*Plus version should appear. If Windows says the command is not recognized, check Path and confirm that `sqlplus.exe` exists in the specified folder.

3. Configure the Autonomous Database wallet

1. Create a folder for the wallet, for example:

```
C:\oracle\wallets\sqlworkshop
```

2. Extract the wallet ZIP into that folder. The files should be directly inside this folder, not hidden one level deeper. Typical files include `tnsnames.ora`, `sqlnet.ora`, and wallet key/certificate files.

3. Open `sqlnet.ora` in Notepad. If it contains a placeholder wallet path, replace that path with the actual wallet folder. Preserve the existing syntax and other settings.

4. Set the Windows **TNS_ADMIN** user environment variable to the folder containing `tnsnames.ora` and `sqlnet.ora`:

```
TNS_ADMIN=C:\oracle\wallets\sqlworkshop
```

To set it: Start → search for “Edit environment variables for your account” → under User variables select New → enter variable name `TNS_ADMIN` and the wallet folder as its value → OK.

5. Close and reopen Command Prompt so the new environment variable is loaded.

> **Security:** A wallet contains sensitive connection material. Share it only through an approved secure channel. Do not upload it to GitHub or include it in public course materials. Follow your administrator's instructions if the wallet ZIP is password-protected.

4. Check the connection alias

Open `tnsnames.ora` from the wallet folder in Notepad. Find the service alias supplied for the training database. It may look like this:

```
sqlworkshop_medium = ...
```

The actual descriptor after the equals sign is generated for your database; do not replace it with the dots shown here. Wallets often include aliases for low, medium, and high service levels. Use an alias that appears exactly in your own `tnsnames.ora`.

5. Connect using SQL*Plus

In a new Command Prompt, run the following command, replacing the username with the one provided to you:

```
sqlplus YOUR_USERNAME@sqlworkshop_medium
```

SQL*Plus prompts for your password. Type it and press Enter; the characters may not appear as you type. This is normal.

If your assigned alias differs, replace `sqlworkshop_medium` with the exact alias in `tnsnames.ora`. Do not put the password directly in the command.

6. Verify the connection

When the connection succeeds, SQL*Plus displays a connected message and the SQL> prompt. Run:

```
SELECT USER FROM dual;

SELECT sys_context('USERENV', 'SERVICE_NAME') AS service_name
FROM dual;
```

The first query shows the database user for the session. The second shows the service name used by the connection. To disconnect, enter:

```
EXIT
```

7. Common problems and fixes

Symptom / error	Checks and likely fix
sqlplus is not recognized	Check that Path points to the Instant Client folder containing sqlplus.exe. Reopen Command Prompt after changing Path.
ORA-12154: connect identifier could not be resolved	Check TNS_ADMIN, confirm tnsnames.ora exists in that folder, and spell the alias exactly as listed.
ORA-12514 / ORA-12541 / timeout	Check the alias, database availability, network/firewall access, and whether the wallet belongs to the intended database.
ORA-01017: invalid username/password	Re-enter credentials carefully. Ask the administrator to verify the account or reset the password if needed.
Wallet or TLS error	Check the wallet folder path in sqlnet.ora and TNS_ADMIN. Ensure the complete wallet ZIP was extracted and files were not renamed or removed.
Connection seems to pause after password entry	Wait briefly; connection time depends on the network and database availability. If it continues for several minutes, check the displayed error and network access.

8. Participant checklist

- ■ Downloaded Instant Client Basic and SQL*Plus ZIPs of the same version.
- ■ Extracted both packages into one Instant Client folder.
- ■ Added that folder to Path and verified with sqlplus -v.
- ■ Extracted the wallet ZIP into a dedicated folder.
- ■ Checked the wallet path in sqlnet.ora, where applicable.
- ■ Set TNS_ADMIN to the wallet folder and reopened Command Prompt.
- ■ Confirmed the exact service alias in tnsnames.ora.
- ■ Connected with sqlplus USERNAME@service_alias and entered the password at the prompt.
- ■ Successfully ran SELECT USER FROM dual;

Official resource

Oracle Instant Client Downloads

Instructor note

Provide participants with the official download link, wallet ZIP through an approved secure channel, database username, and password separately. The alias shown in this guide is an example; participants must use the alias present in their wallet's `tnsnames.ora`.

This guide covers Windows command-line setup for Oracle SQL*Plus with an Autonomous Database wallet. Package names and versions can change; use Oracle's official download page and current administrator instructions.